It is worth noting that $\Lambda(h) = \mathcal{O}(h^2)$, i.e. for small steps Λ will also be small. This justifies the initial guess $\Lambda^{(0)} = 0$, and it also suggests that if convergence of the Newton iteration is poor, this might be improved by taking a smaller timestep, since this effectively places $\Lambda^{(0)}$ closer to the solution of the equation.

In the context of large systems with large numbers of constraints it makes sense to invest significant effort to ensure that the algebraic equations are solved in the most efficient way possible. In [25], this topic was studied in depth, and a variety of iterative Newton-based numerical algorithms were proposed and compared.

4.3.3 *Iterative Component-Wise Solution of the Nonlinear Equations*

An alternative method for solving (4.17) is based on solving for $\boldsymbol{\Lambda}$ *component-wise* [322]. To describe this algorithm let us enumerate the equations as

$$g_j(\boldsymbol{Q}_n - \boldsymbol{M}^{-1}\boldsymbol{G}_n^k \Lambda_k) = 0, \qquad \boldsymbol{G}_n^k = \nabla g_k(\boldsymbol{q}_n),$$

i.e. $\boldsymbol{G}_n^k$ is the kth column of $\boldsymbol{G}_n^T$.

We begin with the initializations $\boldsymbol{\Lambda} = \boldsymbol{0}$ and $\boldsymbol{Q} = \boldsymbol{Q}_n$, and solve for $\Delta\Lambda_1$ from

$$g_1(\boldsymbol{Q} - \boldsymbol{M}^{-1}\boldsymbol{G}_n^1 \Delta\Lambda_1) = 0.$$

This is scalar nonlinear equation for $\Delta\Lambda_1$. We approximate its solution using a linear approximation at $\boldsymbol{Q}$ thus we have

$$\Delta\Lambda_1 = \left[\nabla g_1(\boldsymbol{Q})^T \boldsymbol{M}^{-1}\boldsymbol{G}_n^1\right]^{-1} g_1(\boldsymbol{Q}).$$

We then update Q by

$$\boldsymbol{Q} := \boldsymbol{Q} - \boldsymbol{M}^{-1}\boldsymbol{G}_n^1 \Delta\Lambda_1,$$

and continue to update Λ_2 in the same manner (using the second constraint). In general, we perform a sequence of updates

$$\begin{aligned} \Delta\Lambda_j &= \left[\nabla g_j(\boldsymbol{Q})^T \boldsymbol{M}^{-1}\boldsymbol{G}_j^1\right]^{-1} g_j(\boldsymbol{Q}), \\ \boldsymbol{Q} &:= \boldsymbol{Q} - \boldsymbol{M}^{-1}\boldsymbol{G}_n^j \Delta\Lambda_j, \end{aligned}$$

as j cycles repeatedly from 1 to m. We halt the iteration when an appropriate stopping criterion is satisfied, which might be

$$\|\Delta\Lambda_j\| \le \text{tol}, \qquad j = 1, 2, \ldots, m.$$